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## 31.1 Elementary number-theoretic notions

This section provides a brief review of notions from elementary number theory concerning the set  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$  of integers and the set  $\mathbb{N} = \{0, 1, 2, \dots\}$  of natural numbers.

### Divisibility and divisors

The notion of one integer being divisible by another is key to the theory of numbers. The notation  $d \mid a$  (read “ $d$  **divides**  $a$ ”) means that  $a = kd$  for some integer  $k$ . Every integer divides 0. If  $a > 0$  and  $d \mid a$ , then  $|d| \leq |a|$ . If  $d \mid a$ , then we also say that  $a$  is a **multiple** of  $d$ . If  $d$  does not divide  $a$ , we write  $d \nmid a$ .

If  $d \mid a$  and  $d \geq 0$ , then  $d$  is a **divisor** of  $a$ . Since  $d \mid a$  if and only if  $-d \mid a$ , without loss of generality, we define the divisors of  $a$  to be nonnegative, with the understanding that the negative of any divisor of  $a$  also divides  $a$ . A divisor of a nonzero integer  $a$  is at least 1 but not greater than  $|a|$ . For example, the divisors of 24 are 1, 2, 3, 4, 6, 8, 12, and 24.

Every positive integer  $a$  is divisible by the **trivial divisors** 1 and  $a$ . The nontrivial divisors of  $a$  are the **factors** of  $a$ . For example, the factors of 20 are 2, 4, 5, and 10.

### Prime and composite numbers

An integer  $a > 1$  whose only divisors are the trivial divisors 1 and  $a$  is a **prime number** or, more simply, a **prime**. Primes have many special properties and play a critical role in number theory. The first 20 primes, in order, are

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71.

Exercise 31.1-2 asks you to prove that there are infinitely many primes. An integer  $a > 1$  that is not prime is a **composite number** or, more simply, a **composite**. For example, 39 is composite because  $3 \mid 39$ . We call the integer 1 a **unit**, and it is neither prime nor composite. Similarly, the integer 0 and all negative integers are neither prime nor composite.

## The division theorem, remainders, and modular equivalence

Given an integer  $n$ , we can partition the integers into those that are multiples of  $n$  and those that are not multiples of  $n$ . Much number theory is based upon refining this partition by classifying the integers that are not multiples of  $n$  according to their remainders when divided by  $n$ . The following theorem provides the basis for this refinement. We omit the proof (but see, for example, Niven and Zuckerman [345]).

### *Theorem 31.1 (Division theorem)*

For any integer  $a$  and any positive integer  $n$ , there exist unique integers  $q$  and  $r$  such that  $0 \leq r < n$  and  $a = qn + r$ . ■

The value  $q = \lfloor a/n \rfloor$  is the *quotient* of the division. The value  $r = a \bmod n$  is the *remainder* (or *residue*) of the division, so that  $n \mid a$  if and only if  $a \bmod n = 0$ .

The integers partition into  $n$  equivalence classes according to their remainders modulo  $n$ . The *equivalence class modulo  $n$*  containing an integer  $a$  is

$$[a]_n = \{a + kn : k \in \mathbb{Z}\}.$$

For example,  $[3]_7 = \{\dots, -11, -4, 3, 10, 17, \dots\}$ , and  $[-4]_7$  and  $[10]_7$  also denote this set. With the notation defined on page 64, writing  $a \in [b]_n$  is the same as writing  $a = b \pmod{n}$ . The set of all such equivalence classes is

$$\mathbb{Z}_n = \{[a]_n : 0 \leq a \leq n-1\}. \quad (31.1)$$

When you see the definition

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}, \quad (31.2)$$

you should read it as equivalent to equation (31.1) with the understanding that 0 represents  $[0]_n$ , 1 represents  $[1]_n$ , and so on. Each class is represented by its smallest nonnegative element. You should keep the underlying equivalence classes in mind, however. For example,

if we refer to  $-1$  as a member of  $\mathbb{Z}_n$ , we are really referring to  $[n-1]_n$ , since  $-1 = n-1 \pmod{n}$ .

### Common divisors and greatest common divisors

If  $d$  is a divisor of  $a$  and  $d$  is also a divisor of  $b$ , then  $d$  is a **common divisor** of  $a$  and  $b$ . For example, the divisors of 30 are 1, 2, 3, 5, 6, 10, 15, and 30, and so the common divisors of 24 and 30 are 1, 2, 3, and 6. Any pair of integers has a common divisor of 1.

An important property of common divisors is that

$$\text{if } d \mid a \text{ and } d \mid b, \text{ then } d \mid (a+b) \text{ and } d \mid (a-b). \quad (31.3)$$

More generally, for any integers  $x$  and  $y$ ,

$$\text{if } d \mid a \text{ and } d \mid b, \text{ then } d \mid (ax+by). \quad (31.4)$$

Also, if  $a \mid b$ , then either  $|a| \leq |b|$  or  $b = 0$ , which implies that

$$\text{if } a \mid b \text{ and } b \mid a, \text{ then } a = \pm b. \quad (31.5)$$

The **greatest common divisor** of two integers  $a$  and  $b$  which are not both 0, denoted by  $\gcd(a, b)$ , is the largest of the common divisors of  $a$  and  $b$ . For example,  $\gcd(24, 30) = 6$ ,  $\gcd(5, 7) = 1$ , and  $\gcd(0, 9) = 9$ . If  $a$  and  $b$  are both nonzero, then  $\gcd(a, b)$  is an integer between 1 and  $\min\{|a|, |b|\}$ . We define  $\gcd(0, 0)$  to be 0, so that standard properties of the gcd function (such as equation (31.9) below) hold universally.

Exercise 31.1-9 asks you to prove the following elementary properties of the gcd function:

$$\gcd(a, b) = \gcd(b, a), \quad (31.6)$$

$$\gcd(a, b) = \gcd(-a, b), \quad (31.7)$$

$$\gcd(a, b) = \gcd(|a|, |b|), \quad (31.8)$$

$$\gcd(a, 0) = |a|, \quad (31.9)$$

$$\gcd(a, ka) = |a| \quad \text{for any } k \in \mathbb{Z}. \quad (31.10)$$

The following theorem provides an alternative and useful way to characterize  $\gcd(a, b)$ .

### **Theorem 31.2**

If  $a$  and  $b$  are any integers, not both zero, then  $\gcd(a, b)$  is the smallest positive element of the set  $\{ax + by : x, y \in \mathbb{Z}\}$  of linear combinations of  $a$  and  $b$ .

**Proof** Let  $s$  be the smallest positive such linear combination of  $a$  and  $b$ , and let  $s = ax + by$  for some  $x, y \in \mathbb{Z}$ . Let  $q = \lfloor a/s \rfloor$ . Equation (3.11) on page 64 then implies

$$\begin{aligned} a \bmod s &= a - qs \\ &= a - q(ax + by) \\ &= a(1 - qx) + b(-qy), \end{aligned}$$

so that  $a \bmod s$  is a linear combination of  $a$  and  $b$  as well. Because  $s$  is the smallest *positive* such linear combination and  $0 \leq a \bmod s < s$  (inequality (3.12) on page 64),  $a \bmod s$  cannot be positive. Hence,  $a \bmod s = 0$ . Therefore, we have that  $s \mid a$  and, by analogous reasoning,  $s \mid b$ . Thus,  $s$  is a common divisor of  $a$  and  $b$ , so that  $\gcd(a, b) \geq s$ . By definition,  $\gcd(a, b)$  divides both  $a$  and  $b$ , and  $s$  is defined as a linear combination of  $a$  and  $b$ . Equation (31.4) therefore implies that  $\gcd(a, b) \mid s$ . But  $\gcd(a, b) \mid s$  and  $s > 0$  imply that  $\gcd(a, b) \leq s$ . Combining  $\gcd(a, b) \geq s$  and  $\gcd(a, b) \leq s$  yields  $\gcd(a, b) = s$ . We conclude that  $s$ , the smallest positive linear combination of  $a$  and  $b$ , is also their greatest common divisor. ■

Theorem 31.2 engenders three useful corollaries.

### **Corollary 31.3**

For any integers  $a$  and  $b$ , if  $d \mid a$  and  $d \mid b$ , then  $d \mid \gcd(a, b)$ .

**Proof** This corollary follows from equation (31.4) and Theorem 31.2, because  $\gcd(a, b)$  is a linear combination of  $a$  and  $b$ , ■

### **Corollary 31.4**

For all integers  $a$  and  $b$  and any nonnegative integer  $n$ , we have

$$\gcd(an, bn) = n \gcd(a, b).$$

**Proof** If  $n = 0$ , the corollary is trivial. If  $n > 0$ , then  $\gcd(an, bn)$  is the smallest positive element of the set  $\{anx + bny : x, y \in \mathbb{Z}\}$ , which in turn is  $n$  times the smallest positive element of the set  $\{ax + by : x, y \in \mathbb{Z}\}$ . ■

### **Corollary 31.5**

For all positive integers  $n$ ,  $a$ , and  $b$ , if  $n \mid ab$  and  $\gcd(a, n) = 1$ , then  $n \mid b$ .

**Proof** Exercise 31.1-5 asks you to provide the proof. ■

## **Relatively prime integers**

Two integers  $a$  and  $b$  are *relatively prime* if their only common divisor is 1, that is, if  $\gcd(a, b) = 1$ . For example, 8 and 15 are relatively prime, since the divisors of 8 are 1, 2, 4, and 8, and the divisors of 15 are 1, 3, 5, and 15. The following theorem states that if two integers are each relatively prime to an integer  $p$ , then their product is relatively prime to  $p$ .

### **Theorem 31.6**

For any integers  $a$ ,  $b$ , and  $p$ , we have  $\gcd(ab, p) = 1$  if and only if  $\gcd(a, p) = 1$  and  $\gcd(b, p) = 1$  both hold.

**Proof** If  $\gcd(a, p) = 1$  and  $\gcd(b, p) = 1$ , then it follows from Theorem 31.2 that there exist integers  $x, y, x'$ , and  $y'$  such that

$$ax + py = 1,$$

$$bx' + py' = 1.$$

Multiplying these equations and rearranging gives

$$ab(xx') + p(ybx' + y'ax + pyy') = 1.$$

Since 1 is thus a positive linear combination of  $ab$  and  $p$ , it is the smallest positive linear combination. Applying Theorem 31.2 implies  $\gcd(ab, p) = 1$ , completing the proof in this direction.

Conversely, if  $\gcd(ab, p) = 1$ , then Theorem 31.2 implies that there exist integers  $x$  and  $y$  such that

$$abx + py = 1.$$

Writing  $abx$  as  $a(bx)$  and applying Theorem 31.2 again proves that  $\gcd(a, p) = 1$ . Proving that  $\gcd(b, p) = 1$  is similar. ■

Integers  $n_1, n_2, \dots, n_k$  are *pairwise relatively prime* if  $\gcd(n_i, n_j) = 1$  for  $1 \leq i < j \leq k$ .

### Unique prime factorization

An elementary but important fact about divisibility by primes is the following.

#### **Theorem 31.7**

For all primes  $p$  and all integers  $a$  and  $b$ , if  $p \mid ab$ , then  $p \mid a$  or  $p \mid b$  (or both).

**Proof** Assume for the purpose of contradiction that  $p \mid ab$ , but that  $p \nmid a$  and  $p \nmid b$ . Because  $p > 1$  and  $ab = kp$  for some  $k \in \mathbb{Z}$ , equation (31.10) gives that  $\gcd(ab, p) = p$ . We also have that  $\gcd(a, p) = 1$  and  $\gcd(b, p) = 1$ , since the only divisors of  $p$  are 1 and  $p$ , and we assumed that  $p$  divides neither  $a$  nor  $b$ . Theorem 31.6 then implies that  $\gcd(ab, p) = 1$ , contradicting  $\gcd(ab, p) = p$ . This contradiction completes the proof. ■

A consequence of Theorem 31.7 is that any composite integer can be uniquely factored into a product of primes. Exercise 31.1-11 asks you to provide a proof.

#### **Theorem 31.8 (Unique prime factorization)**

There is exactly one way to write any composite integer  $a$  as a product of the form

$$a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r},$$

where the  $p_i$  are prime,  $p_1 < p_2 < \dots < p_r$ , and the  $e_i$  are positive integers. ■

As an example, the unique prime factorization of the number 6000 is  $2^4 \cdot 3^1 \cdot 5^3$ .

## Exercises

### 31.1-1

Prove that if  $a > b > 0$  and  $c = a + b$ , then  $c \bmod a = b$ .

### 31.1-2

Prove that there are infinitely many primes. (*Hint*: Show that none of the primes  $p_1, p_2, \dots, p_k$  divide  $(p_1 p_2 \cdots p_k) + 1$ .)

### 31.1-3

Prove that if  $a \mid b$  and  $b \mid c$ , then  $a \mid c$ .

### 31.1-4

Prove that if  $p$  is prime and  $0 < k < p$ , then  $\gcd(k, p) = 1$ .

### 31.1-5

Prove Corollary 31.5.

### 31.1-6

Prove that if  $p$  is prime and  $0 < k < p$ , then  $\binom{p}{k}$ . Conclude that for all integers  $a$  and  $b$  and all primes  $p$ ,

$$(a + b)^p = a^p + b^p \pmod{p}.$$

### 31.1-7

Prove that if  $a$  and  $b$  are any positive integers such that  $a \mid b$ , then

$$(x \bmod b) \bmod a = x \bmod a$$

for any  $x$ . Prove, under the same assumptions, that

$$x = y \pmod{b} \text{ implies } x = y \pmod{a}$$

for any integers  $x$  and  $y$ .

### 31.1-8

For any integer  $k > 0$ , an integer  $n$  is a *kth power* if there exists an integer  $a$  such that  $a^k = n$ . Furthermore,  $n > 1$  is a *nontrivial power* if it is a  $k$ th power for some integer  $k > 1$ . Show how to determine whether a given  $\beta$ -bit integer  $n$  is a nontrivial power in time polynomial in  $\beta$ .

**31.1-9**

Prove equations (31.6)–(31.10).

**31.1-10**

Show that the gcd operator is associative. That is, prove that for all integers  $a$ ,  $b$ , and  $c$ , we have

$$\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c).$$

★ **31.1-11**

Prove Theorem 31.8.

**31.1-12**

Give efficient algorithms for the operations of dividing a  $\beta$ -bit integer by a shorter integer and of taking the remainder of a  $\beta$ -bit integer when divided by a shorter integer. Your algorithms should run in  $\Theta(\beta^2)$  time.

**31.1-13**

Give an efficient algorithm to convert a given  $\beta$ -bit (binary) integer to a decimal representation. Argue that if multiplication or division of integers whose length is at most  $\beta$  takes  $M(\beta)$  time, where  $M(\beta) = \Omega(\beta)$ , then you can convert binary to decimal in  $O(M(\beta) \lg \beta)$  time. (*Hint:* Use a divide-and-conquer approach, obtaining the top and bottom halves of the result with separate recursions.)

**31.1-14**

Professor Marshall sets up  $n$  lightbulbs in a row. The lightbulbs all have switches, so that if he presses a bulb, it toggles on if it was off and off if it was on. The lightbulbs all start off. For  $i = 1, 2, 3, \dots, n$ , the professor presses bulb  $i, 2i, 3i, \dots$ . After the last press, which lightbulbs are on? Prove your answer.

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## 31.2 Greatest common divisor

In this section, we describe Euclid's algorithm for efficiently computing the greatest common divisor of two integers. When we analyze the running time, we'll see a surprising connection with the Fibonacci numbers, which yield a worst-case input for Euclid's algorithm.

We restrict ourselves in this section to nonnegative integers. This restriction is justified by equation (31.8), which states that  $\gcd(a, b) = \gcd(|a|, |b|)$ .

In principle, for positive integers  $a$  and  $b$ , their prime factorizations suffice to compute  $\gcd(a, b)$ . Indeed, if

$$a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}, \quad (31.11)$$

$$b = p_1^{f_1} p_2^{f_2} \cdots p_r^{f_r}, \quad (31.12)$$

with 0 exponents being used to make the set of primes  $p_1, p_2, \dots, p_r$  the same for both  $a$  and  $b$ , then, as Exercise 31.2-1 asks you to show,

$$\gcd(a, b) = p_1^{\min\{e_1, f_1\}} p_2^{\min\{e_2, f_2\}} \cdots p_r^{\min\{e_r, f_r\}}. \quad (31.13)$$

The best algorithms to date for factoring do not run in polynomial time. Thus, this approach to computing greatest common divisors seems unlikely to yield an efficient algorithm.

Euclid's algorithm for computing greatest common divisors relies on the following theorem.

### ***Theorem 31.9 (GCD recursion theorem)***

For any nonnegative integer  $a$  and any positive integer  $b$ ,

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

**Proof** We will show that  $\gcd(a, b)$  and  $\gcd(b, a \bmod b)$  divide each other. Since they are both nonnegative, equation (31.5) then implies that they must be equal.

We first show that  $\gcd(a, b) \mid \gcd(b, a \bmod b)$ . If we let  $d = \gcd(a, b)$ , then  $d \mid a$  and  $d \mid b$ . By equation (3.11) on page 64,  $a \bmod b = a - qb$ , where  $q = \lfloor a/b \rfloor$ . Since  $a \bmod b$  is thus a linear combination of  $a$  and  $b$ , equation (31.4) implies that  $d \mid (a \bmod b)$ . Therefore, since  $d \mid b$  and  $d \mid (a \bmod b)$ , Corollary 31.3 implies that  $d \mid \gcd(b, a \bmod b)$ , that is,